# GTIIT Spring 2025 Calculus 1M
## **Workshop 3** - Tuesday

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Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

**If you need any help at all —whether it's a tiny hint, a big clue, or the full solution— don't hesitate to ask me!**

***Important note:*** If you find any errors or typos, please let me know so I can correct them.

1. Find an expression for a cubic function $f$ if $f(1) = 6$ and $f(-1) = f(0) = f(2) = 0$.
2. Find $f \circ g \circ h$ of the following functions:
   (a) $f(x) = x^2$, $g(x) = x^2 + 1$, $h(x) = x - 1$
   (b) $f(x) = 3x - 2$, $g(x) = \sin x$, $h(x) = x^2$
3. Suppose that $g$ is an even function and let $h = f \circ g$. Is $h$ even, odd, or neither?
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5. Starting with the function $f(x) = e^x$, write down the equation of the graph that result from:
   (a) reflecting the graph of $f$ about the line $y = 3$,
   (b) reflecting the graph of $f$ about the line $x = 1$.
6. Find the domain of the function $f(x) = \frac{1-e^{x^2}}{1-e^{1-x}}$.
7. Find the domain of $f(x) = \ln(e^x - 4)$. Compute $f^{-1}$ and find its domain.
8. Find the domain and range of the function
$$g(x) = \arcsin(3x + 1).$$
9. Evaluate the following limits using the basic limit properties:
   (a) $\lim_{x\to2}(3x^2 - 2x + 1)$
   (b) $\lim_{x\to3} \frac{\frac{1}{x}-\frac{1}{3}}{x-3}$
   (c) $\lim_{h\to0} \frac{\frac{1}{(x+h)^2}-\frac{1}{x^2}}{h}$

10. Determine the limits $\lim_{x\to 5^-} \frac{x+1}{x-5}$ and $\lim_{x\to 5^+} \frac{x+1}{x-5}$

11. Find the vertical asymptotes of the function:

$$y = \frac{x^2+1}{3x-2x^2}$$

12. **Application of the Squeeze Theorem:**
Define the function

$$f(x) = x^2 \cos\left(\frac{1}{x}\right) \sin\left(\frac{1}{x}\right) \quad \text{for } x \neq 0,$$

and set $f(0) = 0$. Use the squeeze theorem to prove that

$$\lim_{x\to 0} f(x) = 0.$$

*Hint:* Find appropriate functions $g(x)$ and $h(x)$ that bound $f(x)$ from below and above, respectively.

13. **Absolute Value and Limit Existence:**
Construct a function $f : \mathbb{R} \to \mathbb{R}$ such that the limit $\lim_{x\to 0} f(x)$ does not exist, but the limit $\lim_{x\to 0} |f(x)|$ exists. Provide a rigorous justification for your example.

14. **Lateral Limits, Absolute Value, and Overall Limit:**
Consider the function

$$f(x) = \begin{cases} \frac{|x|}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

(a) Determine $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^+} f(x)$.

(b) Explain why the overall limit $\lim_{x\to 0} f(x)$ does not exist, and compute $\lim_{x\to 0} |f(x)|$.